

## 6.2. Effect of GOP numbers

Table 5: Ablation study of GOP numbers on MSR-VTT test subset.

| KeyInt | GOP Nums<br>(\$N\$) | Percentile       | Inference Time<br>on Test Set (s) | B4           | M            | R            | C            |
|--------|---------------------|------------------|-----------------------------------|--------------|--------------|--------------|--------------|
| 40     | 9                   | 25 <sup>th</sup> | 405.96                            | 48.37        | 31.18        | 65.22        | 63.46        |
| 40     | 10                  | 50 <sup>th</sup> | 407.92                            | 48.10        | 31.25        | 64.97        | 63.86        |
| 40     | 13                  | 75 <sup>th</sup> | 411.46                            | <b>48.54</b> | <b>31.30</b> | 65.16        | <b>64.29</b> |
| 40     | 16                  | 85 <sup>th</sup> | 414.40                            | 48.37        | 31.28        | <b>65.27</b> | 64.08        |
| 40     | 20                  | 95 <sup>th</sup> | 418.04                            | 48.07        | 31.23        | 65.12        | 63.77        |

## 6.3. Effect of the number of stacked decoder layers

Table 6: Ablation study of module layers on the MSR-VTT test subset. Dec\_BD and Dec\_FD denote the number of stacked Transformer layers in the Backward Decoder and Forward Decoder, respectively. #Params denotes the total number of trainable parameters in the model.

| Dec_BD | Dec_FD | #Params | B4           | M            | R            | C            |
|--------|--------|---------|--------------|--------------|--------------|--------------|
| 1      | 1      | 34.2M   | 48.69        | 31.35        | <b>65.41</b> | 62.77        |
| 2      | 2      | 43.6M   | 48.75        | <b>31.50</b> | 65.05        | 63.37        |
| 3      | 3      | 53.1M   | 48.54        | 31.30        | 65.16        | <b>64.29</b> |
| 4      | 4      | 62.6M   | <b>48.92</b> | 31.23        | 65.27        | 62.83        |
| 5      | 5      | 72.0M   | 47.95        | 31.28        | 65.07        | 61.90        |
| 6      | 6      | 81.5M   | 48.68        | 31.46        | 65.31        | 62.23        |